

A Detailed LaTeX Math Document on Maxwell's Equations

Integral and Differential Forms, Derivations, Names, Physical Meaning, and Main Consequences

Contents

1	Introduction	4
2	Notation and Preliminaries	4
2.1	Divergence Theorem	5
2.2	Stokes' Theorem	5
2.3	A Key Vector Identity	5
3	The Four Maxwell Equations: Names and Forms	5
3.1	Gauss's Law for Electricity	5
3.2	Gauss's Law for Magnetism	5
3.3	Faraday's Law of Induction	6
3.4	Ampère-Maxwell Law	6
4	A Summary Table	6
5	Gauss's Law for Electricity	6
5.1	Physical Meaning	6
5.2	From Coulomb's Law to the Integral Form	6
5.3	Deriving the Differential Form from the Integral Form	7
5.4	Deriving the Integral Form from the Differential Form	7
6	Gauss's Law for Magnetism	8
6.1	Physical Meaning	8
6.2	Differential Form from Integral Form	8
6.3	Integral Form from Differential Form	8
6.4	Important Consequence	8
7	Faraday's Law of Induction	8

7.1	Physical Meaning	8
7.2	Integral Form	9
7.3	Differential Form from Integral Form	9
7.4	Integral Form from Differential Form	9
8	Ampère's Law and Maxwell's Correction	10
8.1	Original Ampère's Law	10
8.2	Why the Original Form Fails	10
8.3	Maxwell's Correction: The Displacement Current Term	10
8.4	Integral Form	11
8.5	Differential Form from Integral Form	11
9	Charge Conservation as a Consequence of Maxwell's Equations	12
10	Why Integral and Differential Forms Mean Different Things Physically	12
10.1	Integral Form	12
10.2	Differential Form	12
11	Electromagnetic Waves in Vacuum	13
11.1	Wave Equation for the Electric Field	13
11.2	Wave Equation for the Magnetic Field	13
11.3	Plane Wave Structure	14
12	Potentials and the Homogeneous Maxwell Equations	14
12.1	Vector Potential	14
12.2	Scalar Potential	14
12.3	Gauge Freedom	15
13	Maxwell's Equations in Matter	15
13.1	Simple Linear Media	16
14	Boundary Conditions from Maxwell's Equations	16
14.1	Normal Component of $\mathbf{E}$	16
14.2	Normal Component of $\mathbf{B}$	16
14.3	Tangential Component of $\mathbf{E}$	16

14.4 Tangential Component of $\mathbf{B}$	17
15 Energy Conservation and the Poynting Theorem	17
16 Momentum and the Lorentz Force	18
17 Static Limits	18
17.1 Electrostatics	18
17.2 Magnetostatics	18
18 A Note on “Proofs” Versus “Physical Laws”	19
19 Covariant Form (Advanced but Very Important)	19
19.1 Inhomogeneous Equations	20
19.2 Homogeneous Equations	20
20 Conceptual Interpretation of Each Equation	20
20.1 Gauss's Law for Electricity	20
20.2 Gauss's Law for Magnetism	20
20.3 Faraday's Law	20
20.4 Ampère-Maxwell Law	20
21 A Very Short Logical Summary	21
22 Conclusion	21

1 Introduction

Maxwell's equations are the fundamental field equations of classical electromagnetism. They describe how electric fields $\mathbf{E}$ and magnetic fields $\mathbf{B}$ are generated by charge density ρ and current density $\mathbf{J}$, and how these fields evolve in time.

They can be written in two equivalent classical forms:

- **Integral form**, which directly relates fields to fluxes and circulations over finite surfaces and loops.
- **Differential form**, which gives the local point-by-point relations between the fields and their sources.

It is important to be precise about the word *derivation*. In physics, Maxwell's equations are not *proved* from pure mathematics alone. They are empirical laws, distilled from experiment and symmetry. What *can* be mathematically proved are:

1. the equivalence between the integral and differential forms,
2. the necessity of some terms from consistency conditions,
3. many important consequences such as charge conservation, electromagnetic waves, boundary conditions, energy conservation, and potentials.

In this document we work mainly in **SI units**.

2 Notation and Preliminaries

We denote:

$$\begin{aligned}\mathbf{E}(\mathbf{r}, t) &:= \text{electric field,} \\ \mathbf{B}(\mathbf{r}, t) &:= \text{magnetic field,} \\ \rho(\mathbf{r}, t) &:= \text{charge density,} \\ \mathbf{J}(\mathbf{r}, t) &:= \text{current density,} \\ \varepsilon_0 &:= \text{vacuum permittivity,} \\ \mu_0 &:= \text{vacuum permeability.}\end{aligned}$$

The speed of light in vacuum is

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}. \tag{1}$$

We will use three central vector calculus theorems:

2.1 Divergence Theorem

For a vector field $\mathbf{F}$,

$$\iiint_V (\nabla \cdot \mathbf{F}) dV = \oint_{\partial V} \mathbf{F} \cdot d\mathbf{a}. \quad (2)$$

2.2 Stokes' Theorem

For a vector field $\mathbf{F}$,

$$\oint_{\partial S} \mathbf{F} \cdot d\boldsymbol{\ell} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{a}. \quad (3)$$

2.3 A Key Vector Identity

$$\nabla \times (\nabla \times \mathbf{F}) = \nabla(\nabla \cdot \mathbf{F}) - \nabla^2 \mathbf{F}. \quad (4)$$

Also,

$$\nabla \cdot (\nabla \times \mathbf{F}) = 0. \quad (5)$$

3 The Four Maxwell Equations: Names and Forms

In vacuum, the four Maxwell equations are:

3.1 Gauss's Law for Electricity

Integral form:

$$\oint_S \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\epsilon_0}. \quad (6)$$

Differential form:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}. \quad (7)$$

3.2 Gauss's Law for Magnetism

Integral form:

$$\oint_S \mathbf{B} \cdot d\mathbf{a} = 0. \quad (8)$$

Differential form:

$$\nabla \cdot \mathbf{B} = 0. \quad (9)$$

3.3 Faraday's Law of Induction

Integral form:

$$\oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d}{dt} \iint_S \mathbf{B} \cdot d\mathbf{a}. \quad (10)$$

Differential form:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \quad (11)$$

3.4 Ampère-Maxwell Law

Integral form:

$$\oint_C \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I_{\text{enc}} + \mu_0 \varepsilon_0 \frac{d}{dt} \iint_S \mathbf{E} \cdot d\mathbf{a}. \quad (12)$$

Differential form:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \quad (13)$$

4 A Summary Table

Name	Integral form	Differential form
Gauss's law (electric)	$\oint_S \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\varepsilon_0}$	$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}$
Gauss's law (magnetic)	$\oint_S \mathbf{B} \cdot d\mathbf{a} = 0$	$\nabla \cdot \mathbf{B} = 0$
Faraday's law	$\oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d}{dt} \iint_S \mathbf{B} \cdot d\mathbf{a}$	$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$
Ampère-Maxwell law	$\oint_C \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I_{\text{enc}} + \mu_0 \varepsilon_0 \frac{d}{dt} \iint_S \mathbf{E} \cdot d\mathbf{a}$	$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}$

5 Gauss's Law for Electricity

5.1 Physical Meaning

Gauss's law for electricity says that electric charge is a source (or sink) of electric field. Positive charge produces outward electric flux, negative charge produces inward electric flux.

5.2 From Coulomb's Law to the Integral Form

For a point charge q at the origin,

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}. \quad (14)$$

Take a sphere of radius r centered on the charge. On the sphere, $\mathbf{E}$ is radial and constant in magnitude:

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}. \quad (15)$$

Thus the electric flux through the sphere is

$$\Phi_E = \oint_S \mathbf{E} \cdot d\mathbf{a} = E_S da \quad (16)$$

$$= \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} (4\pi r^2) = \frac{q}{\epsilon_0}. \quad (17)$$

This is already Gauss's law for a point charge:

$$\oint_S \mathbf{E} \cdot d\mathbf{a} = \frac{q}{\epsilon_0}. \quad (18)$$

By superposition, for many charges or a continuous charge distribution,

$$\oint_S \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\epsilon_0}. \quad (19)$$

5.3 Deriving the Differential Form from the Integral Form

Suppose Gauss's law holds for every closed surface:

$$\oint_{\partial V} \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} \iiint_V \rho dV. \quad (20)$$

Using the divergence theorem,

$$\iiint_V (\nabla \cdot \mathbf{E}) dV = \frac{1}{\epsilon_0} \iiint_V \rho dV. \quad (21)$$

Therefore

$$\iiint_V \left(\nabla \cdot \mathbf{E} - \frac{\rho}{\epsilon_0} \right) dV = 0 \quad (22)$$

for every volume V , which implies

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}. \quad (23)$$

5.4 Deriving the Integral Form from the Differential Form

Assume

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}. \quad (24)$$

Integrate over a volume V :

$$\iiint_V \nabla \cdot \mathbf{E} dV = \frac{1}{\epsilon_0} \iiint_V \rho dV. \quad (25)$$

Apply the divergence theorem:

$$\oint_{\partial V} \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{\text{enc}}}{\epsilon_0}. \quad (26)$$

Thus the two forms are equivalent.

6 Gauss's Law for Magnetism

6.1 Physical Meaning

Gauss's law for magnetism states that the net magnetic flux through any closed surface is zero:

$$\oint_S \mathbf{B} \cdot d\mathbf{a} = 0. \quad (27)$$

This expresses the empirical fact that isolated magnetic monopoles have not been observed in classical electromagnetism. Magnetic field lines do not begin or end; they form closed loops or extend to infinity in such a way that net flux through a closed surface vanishes.

6.2 Differential Form from Integral Form

If for every closed surface,

$$\oint_{\partial V} \mathbf{B} \cdot d\mathbf{a} = 0, \quad (28)$$

then by the divergence theorem,

$$\iiint_V \nabla \cdot \mathbf{B} \, dV = 0 \quad (29)$$

for every volume V . Hence

$$\nabla \cdot \mathbf{B} = 0. \quad (30)$$

6.3 Integral Form from Differential Form

Conversely, if

$$\nabla \cdot \mathbf{B} = 0, \quad (31)$$

then for every volume V ,

$$\oint_{\partial V} \mathbf{B} \cdot d\mathbf{a} = \iiint_V \nabla \cdot \mathbf{B} \, dV = 0. \quad (32)$$

6.4 Important Consequence

Because $\nabla \cdot \mathbf{B} = 0$, one may define a vector potential $\mathbf{A}$ such that

$$\mathbf{B} = \nabla \times \mathbf{A}. \quad (33)$$

This is guaranteed, at least locally, because every divergence-free field can be represented as the curl of another vector field.

7 Faraday's Law of Induction

7.1 Physical Meaning

Faraday's law says that a time-varying magnetic flux induces circulation of the electric field. In words:

Changing magnetic flux produces an electric field.

This is the basis of transformers, generators, inductors, and electromagnetic induction.

7.2 Integral Form

For a fixed contour C bounding a surface S ,

$$\oint_C \mathbf{E} \cdot d\ell = -\frac{d}{dt} \iint_S \mathbf{B} \cdot d\mathbf{a}. \quad (34)$$

The minus sign is Lenz's law: the induced field opposes the change in flux.

7.3 Differential Form from Integral Form

Apply Stokes' theorem:

$$\oint_C \mathbf{E} \cdot d\ell = \iint_S (\nabla \times \mathbf{E}) \cdot d\mathbf{a}. \quad (35)$$

Hence

$$\iint_S (\nabla \times \mathbf{E}) \cdot d\mathbf{a} = -\frac{d}{dt} \iint_S \mathbf{B} \cdot d\mathbf{a}. \quad (36)$$

For a fixed surface, move the time derivative inside:

$$\iint_S (\nabla \times \mathbf{E}) \cdot d\mathbf{a} = \iint_S \left(-\frac{\partial \mathbf{B}}{\partial t} \right) \cdot d\mathbf{a}. \quad (37)$$

Since this holds for arbitrary surfaces,

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \quad (38)$$

7.4 Integral Form from Differential Form

Start with

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \quad (39)$$

Integrate over a surface S :

$$\iint_S (\nabla \times \mathbf{E}) \cdot d\mathbf{a} = -\iint_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{a}. \quad (40)$$

Apply Stokes' theorem:

$$\oint_{\partial S} \mathbf{E} \cdot d\ell = -\frac{d}{dt} \iint_S \mathbf{B} \cdot d\mathbf{a}, \quad (41)$$

for a fixed surface.

8 Ampère's Law and Maxwell's Correction

8.1 Original Ampère's Law

For steady currents, Ampère found

$$\oint_C \mathbf{B} \cdot d\ell = \mu_0 I_{\text{enc}}, \quad (42)$$

or locally,

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}. \quad (43)$$

This works for magnetostatics, but not in general time-dependent situations.

8.2 Why the Original Form Fails

Take the divergence of the magnetostatic Ampère law:

$$\nabla \cdot (\nabla \times \mathbf{B}) = \mu_0 \nabla \cdot \mathbf{J}. \quad (44)$$

But the divergence of a curl is identically zero, so

$$0 = \mu_0 \nabla \cdot \mathbf{J}, \quad (45)$$

meaning

$$\nabla \cdot \mathbf{J} = 0. \quad (46)$$

This is only valid for steady currents. In general, charge conservation requires the continuity equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0. \quad (47)$$

Therefore, the equation

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} \quad (48)$$

cannot be correct when $\frac{\partial \rho}{\partial t} \neq 0$.

8.3 Maxwell's Correction: The Displacement Current Term

From Gauss's law,

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}. \quad (49)$$

Differentiate with respect to time:

$$\nabla \cdot \frac{\partial \mathbf{E}}{\partial t} = \frac{1}{\varepsilon_0} \frac{\partial \rho}{\partial t}. \quad (50)$$

Use the continuity equation,

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot \mathbf{J}, \quad (51)$$

so

$$\nabla \cdot \left(\mathbf{J} + \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) = 0. \quad (52)$$

This suggests the corrected law

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \quad (53)$$

Now taking the divergence of both sides gives

$$0 = \mu_0 \nabla \cdot \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial}{\partial t}, \quad (54)$$

which becomes

$$0 = \mu_0 \nabla \cdot \mathbf{J} + \mu_0 \frac{\partial \rho}{\partial t}, \quad (55)$$

hence

$$\nabla \cdot \mathbf{J} + \frac{\partial \rho}{\partial t} = 0. \quad (56)$$

So Maxwell's added term restores consistency with charge conservation.

8.4 Integral Form

The corrected Ampère-Maxwell law is

$$\oint_C \mathbf{B} \cdot d\ell = \mu_0 I_{\text{enc}} + \mu_0 \varepsilon_0 \frac{d}{dt} \iint_S \mathbf{E} \cdot d\mathbf{a}. \quad (57)$$

Here

$$I_{\text{enc}} = \iint_S \mathbf{J} \cdot d\mathbf{a}. \quad (58)$$

The term

$$\varepsilon_0 \frac{d}{dt} \iint_S \mathbf{E} \cdot d\mathbf{a} \quad (59)$$

is called the **displacement current**.

8.5 Differential Form from Integral Form

Use Stokes' theorem:

$$\oint_C \mathbf{B} \cdot d\ell = \iint_S (\nabla \times \mathbf{B}) \cdot d\mathbf{a}. \quad (60)$$

Then

$$\iint_S (\nabla \times \mathbf{B}) \cdot d\mathbf{a} = \mu_0 \iint_S \mathbf{J} \cdot d\mathbf{a} + \mu_0 \varepsilon_0 \iint_S \frac{\partial \mathbf{E}}{\partial t} \cdot d\mathbf{a}. \quad (61)$$

Since the surface is arbitrary,

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \quad (62)$$

9 Charge Conservation as a Consequence of Maxwell's Equations

One of the most important consistency checks is that Maxwell's equations imply local conservation of electric charge.

Take the divergence of the Ampère-Maxwell law:

$$\nabla \cdot (\nabla \times \mathbf{B}) = \mu_0 \nabla \cdot \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial}{\partial t}. \quad (63)$$

Because $\nabla \cdot (\nabla \times \mathbf{B}) = 0$, we get

$$0 = \mu_0 \nabla \cdot \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial}{\partial t} \left(\frac{\rho}{\varepsilon_0} \right). \quad (64)$$

Thus

$$\nabla \cdot \mathbf{J} + \frac{\partial \rho}{\partial t} = 0. \quad (65)$$

This is the **continuity equation**.

Remark 1. *The continuity equation says that charge cannot be created or destroyed locally. If charge decreases in a volume, it must be flowing out through the boundary.*

10 Why Integral and Differential Forms Mean Different Things Physically

10.1 Integral Form

The integral form describes *global* statements over finite regions:

- flux through a closed surface,
- circulation around a loop,
- enclosed charge or current.

These are often most useful in highly symmetric problems.

10.2 Differential Form

The differential form is *local*. It tells us what happens at a point:

- divergence measures local source strength,
- curl measures local circulation tendency,
- time derivatives show local dynamical evolution.

11 Electromagnetic Waves in Vacuum

One of Maxwell's greatest achievements was showing that his equations predict self-propagating waves.

Assume vacuum with no free charges and no currents:

$$\rho = 0, \quad \mathbf{J} = 0. \quad (66)$$

Then Maxwell's equations become

$$\nabla \cdot \mathbf{E} = 0, \quad (67)$$

$$\nabla \cdot \mathbf{B} = 0, \quad (68)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad (69)$$

$$\nabla \times \mathbf{B} = \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \quad (70)$$

11.1 Wave Equation for the Electric Field

Take the curl of Faraday's law:

$$\nabla \times (\nabla \times \mathbf{E}) = -\frac{\partial}{\partial t} \nabla \times \mathbf{B}. \quad (71)$$

Use Ampère-Maxwell in vacuum:

$$\nabla \times (\nabla \times \mathbf{E}) = -\mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}. \quad (72)$$

Now use the identity

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}. \quad (73)$$

Since $\nabla \cdot \mathbf{E} = 0$ in vacuum,

$$-\nabla^2 \mathbf{E} = -\mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}. \quad (74)$$

Therefore,

$$\nabla^2 \mathbf{E} - \mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0. \quad (75)$$

This is a wave equation with speed

$$v = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = c. \quad (76)$$

11.2 Wave Equation for the Magnetic Field

Similarly, take the curl of the Ampère-Maxwell law in vacuum:

$$\nabla \times (\nabla \times \mathbf{B}) = \mu_0 \varepsilon_0 \frac{\partial}{\partial t} \nabla \times \mathbf{E}. \quad (77)$$

Use Faraday's law:

$$\nabla \times (\nabla \times \mathbf{B}) = -\mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2}. \quad (78)$$

Since $\nabla \cdot \mathbf{B} = 0$,

$$\nabla^2 \mathbf{B} - \mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2} = 0. \quad (79)$$

Thus $\mathbf{B}$ also propagates as a wave at speed c .

11.3 Plane Wave Structure

A plane electromagnetic wave traveling in direction $\mathbf{k}$ has:

- $\mathbf{E} \perp \mathbf{B}$,
- $\mathbf{E} \perp \mathbf{k}$,
- $\mathbf{B} \perp \mathbf{k}$.

Moreover,

$$B = \frac{E}{c} \quad (80)$$

in vacuum.

12 Potentials and the Homogeneous Maxwell Equations

The two equations

$$\nabla \cdot \mathbf{B} = 0, \quad (81)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad (82)$$

are sometimes called the **homogeneous Maxwell equations**.

12.1 Vector Potential

Since $\nabla \cdot \mathbf{B} = 0$, we can write

$$\mathbf{B} = \nabla \times \mathbf{A}. \quad (83)$$

12.2 Scalar Potential

Insert this into Faraday's law:

$$\nabla \times \mathbf{E} = -\frac{\partial}{\partial t} \mathbf{B} = -\nabla \times \frac{\partial \mathbf{A}}{\partial t}. \quad (84)$$

Hence

$$\nabla \times \left(\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} \right) = 0. \quad (85)$$

A curl-free field is a gradient, so

$$\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} = -\nabla \phi, \quad (86)$$

which gives

$$\mathbf{E} = -\nabla \phi - \frac{\partial \mathbf{A}}{\partial t}. \quad (87)$$

Thus the fields can be expressed in terms of potentials:

$$\boxed{\mathbf{B} = \nabla \times \mathbf{A}, \quad \mathbf{E} = -\nabla \phi - \frac{\partial \mathbf{A}}{\partial t}.} \quad (88)$$

12.3 Gauge Freedom

The potentials are not unique. If $\chi(\mathbf{r}, t)$ is any scalar function, then

$$\mathbf{A}' = \mathbf{A} + \nabla \chi, \quad (89)$$

$$\phi' = \phi - \frac{\partial \chi}{\partial t}, \quad (90)$$

leave $\mathbf{E}$ and $\mathbf{B}$ unchanged.

This is called **gauge invariance**.

13 Maxwell's Equations in Matter

In material media, one often introduces the auxiliary fields $\mathbf{D}$ and $\mathbf{H}$:

$$\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P}, \quad (91)$$

$$\mathbf{H} = \frac{1}{\mu_0} \mathbf{B} - \mathbf{M}, \quad (92)$$

where $\mathbf{P}$ is polarization and $\mathbf{M}$ is magnetization.

Then the macroscopic Maxwell equations become

$$\nabla \cdot \mathbf{D} = \rho_f, \quad (93)$$

$$\nabla \cdot \mathbf{B} = 0, \quad (94)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad (95)$$

$$\nabla \times \mathbf{H} = \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t}, \quad (96)$$

where ρ_f and $\mathbf{J}_f$ are free charge and free current.

Their integral forms are

$$\oint_S \mathbf{D} \cdot d\mathbf{a} = Q_{f,\text{enc}}, \quad (97)$$

$$\oint_S \mathbf{B} \cdot d\mathbf{a} = 0, \quad (98)$$

$$\oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d}{dt} \iint_S \mathbf{B} \cdot d\mathbf{a}, \quad (99)$$

$$\oint_C \mathbf{H} \cdot d\boldsymbol{\ell} = I_{f,\text{enc}} + \frac{d}{dt} \iint_S \mathbf{D} \cdot d\mathbf{a}. \quad (100)$$

13.1 Simple Linear Media

In linear, isotropic, homogeneous media,

$$\mathbf{D} = \varepsilon \mathbf{E}, \quad \mathbf{B} = \mu \mathbf{H}. \quad (101)$$

Then electromagnetic waves propagate with speed

$$v = \frac{1}{\sqrt{\mu\varepsilon}}. \quad (102)$$

14 Boundary Conditions from Maxwell's Equations

Maxwell's equations imply jump conditions across interfaces.

14.1 Normal Component of $\mathbf{E}$

Take a thin pillbox crossing a boundary. Gauss's law gives

$$\hat{\mathbf{n}} \cdot (\mathbf{E}_2 - \mathbf{E}_1) = \frac{\sigma}{\varepsilon_0}, \quad (103)$$

where σ is surface charge density.

14.2 Normal Component of $\mathbf{B}$

Using Gauss's law for magnetism on a pillbox,

$$\hat{\mathbf{n}} \cdot (\mathbf{B}_2 - \mathbf{B}_1) = 0. \quad (104)$$

14.3 Tangential Component of $\mathbf{E}$

Apply Faraday's law to a tiny rectangular loop straddling the boundary:

$$\hat{\mathbf{n}} \times (\mathbf{E}_2 - \mathbf{E}_1) = \mathbf{0}. \quad (105)$$

So tangential $\mathbf{E}$ is continuous if the magnetic flux term vanishes in the infinitesimal limit.

14.4 Tangential Component of $\mathbf{B}$

Apply the Ampère-Maxwell law to a tiny loop:

$$\hat{\mathbf{n}} \times (\mathbf{B}_2 - \mathbf{B}_1) = \mu_0 \mathbf{K}, \quad (106)$$

where $\mathbf{K}$ is surface current density.

15 Energy Conservation and the Poynting Theorem

Maxwell's equations also imply conservation of electromagnetic energy.

Start from

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}, \quad (107)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \quad (108)$$

Dot the first with $\mathbf{E}$:

$$\mathbf{E} \cdot (\nabla \times \mathbf{B}) = \mu_0 \mathbf{E} \cdot \mathbf{J} + \mu_0 \varepsilon_0 \mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t}. \quad (109)$$

Dot the second with $\mathbf{B}/\mu_0$:

$$\frac{1}{\mu_0} \mathbf{B} \cdot (\nabla \times \mathbf{E}) = -\frac{1}{\mu_0} \mathbf{B} \cdot \frac{\partial \mathbf{B}}{\partial t}. \quad (110)$$

Subtract the second equation from the first and use the identity

$$\nabla \cdot (\mathbf{E} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{E}) - \mathbf{E} \cdot (\nabla \times \mathbf{B}). \quad (111)$$

This yields

$$\frac{\partial}{\partial t} \left(\frac{\varepsilon_0}{2} E^2 + \frac{1}{2\mu_0} B^2 \right) + \nabla \cdot \left(\frac{1}{\mu_0} \mathbf{E} \times \mathbf{B} \right) = -\mathbf{J} \cdot \mathbf{E}. \quad (112)$$

Define

$$u = \frac{\varepsilon_0}{2} E^2 + \frac{1}{2\mu_0} B^2 \quad (\text{electromagnetic energy density}), \quad (113)$$

$$\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B} \quad (\text{Poynting vector}). \quad (114)$$

Then

$$\boxed{\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}.} \quad (115)$$

This is the **Poynting theorem**. It says:

- electromagnetic energy in a region can change,
- energy can flow in or out through the boundary via $\mathbf{S}$,
- energy can also be transferred to matter via $\mathbf{J} \cdot \mathbf{E}$.

16 Momentum and the Lorentz Force

Although not itself one of Maxwell's equations, the force law that couples fields to matter is the Lorentz force:

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (116)$$

For a continuous charge distribution, the force density is

$$\mathbf{f} = \rho\mathbf{E} + \mathbf{J} \times \mathbf{B}. \quad (117)$$

This is how charges respond to the fields generated by Maxwell's equations.

17 Static Limits

17.1 Electrostatics

If fields are time-independent and currents are absent,

$$\frac{\partial \mathbf{E}}{\partial t} = 0, \quad \frac{\partial \mathbf{B}}{\partial t} = 0, \quad \mathbf{J} = 0. \quad (118)$$

Then Maxwell's equations reduce to

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}, \quad (119)$$

$$\nabla \times \mathbf{E} = 0. \quad (120)$$

Since $\nabla \times \mathbf{E} = 0$, we have

$$\mathbf{E} = -\nabla\phi, \quad (121)$$

and Gauss's law becomes Poisson's equation:

$$\nabla^2\phi = -\frac{\rho}{\varepsilon_0}. \quad (122)$$

If $\rho = 0$, this becomes Laplace's equation:

$$\nabla^2\phi = 0. \quad (123)$$

17.2 Magnetostatics

If the fields are time-independent,

$$\nabla \cdot \mathbf{B} = 0, \quad (124)$$

$$\nabla \times \mathbf{B} = \mu_0\mathbf{J}. \quad (125)$$

Then the magnetic field is generated by steady currents.

18 A Note on “Proofs” Versus “Physical Laws”

There are different levels of mathematical certainty here:

What is empirical?

The following are physical laws inferred from experiment:

- electric charge sources electric field,
- no magnetic monopoles have been observed,
- changing magnetic flux induces electric circulation,
- currents and changing electric fields generate magnetic circulation.

What is mathematically proved once the laws are assumed?

- equivalence of integral and differential forms,
- charge conservation,
- existence of wave equations,
- existence of potentials for the homogeneous equations,
- boundary conditions,
- energy conservation in the Poynting form.

So the deeper structure is:

$$\textit{experiment} \longrightarrow \textit{field equations} \longrightarrow \textit{mathematical consequences}.$$

19 Covariant Form (Advanced but Very Important)

Maxwell's equations become especially elegant in special relativity.

Define spacetime coordinates

$$x^\mu = (ct, \mathbf{r}), \quad (126)$$

the four-current

$$J^\mu = (c\rho, \mathbf{J}), \quad (127)$$

and the electromagnetic field tensor

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix}. \quad (128)$$

Then Maxwell's equations become:

19.1 Inhomogeneous Equations

$$\partial_\mu F^{\mu\nu} = \mu_0 J^\nu. \quad (129)$$

These encode:

- Gauss's law for electricity,
- Ampère-Maxwell law.

19.2 Homogeneous Equations

$$\partial_\alpha F_{\beta\gamma} + \partial_\beta F_{\gamma\alpha} + \partial_\gamma F_{\alpha\beta} = 0. \quad (130)$$

These encode:

- Gauss's law for magnetism,
- Faraday's law.

This compact formulation shows that electric and magnetic fields are not fundamentally separate objects, but different parts of one relativistic field.

20 Conceptual Interpretation of Each Equation

20.1 Gauss's Law for Electricity

Charge is the source of electric field.

20.2 Gauss's Law for Magnetism

There are no isolated magnetic sources in classical electromagnetism.

20.3 Faraday's Law

Changing magnetic fields create electric fields.

20.4 Ampère-Maxwell Law

Electric currents and changing electric fields create magnetic fields.

21 A Very Short Logical Summary

The logical architecture of Maxwell's theory can be summarized as follows:

1. Charges produce $\mathbf{E}$:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}.$$

2. No magnetic monopoles:

$$\nabla \cdot \mathbf{B} = 0.$$

3. Time-varying $\mathbf{B}$ produces circulating $\mathbf{E}$:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}.$$

4. Currents and time-varying $\mathbf{E}$ produce circulating $\mathbf{B}$:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}.$$

Together, these imply:

- charge conservation,
- electromagnetic waves,
- existence of scalar and vector potentials,
- energy transport by the Poynting vector,
- compatibility with special relativity.

22 Conclusion

Maxwell's equations are one of the greatest syntheses in physics. They unify:

- electrostatics,
- magnetostatics,
- induction,
- optics,
- wave propagation,
- charge conservation,
- and, through their covariant form, special relativity.

Their integral forms describe global statements about flux and circulation; their differential forms describe local field behavior. Their status is partly empirical and partly mathematical: experiment gives the laws, while mathematics reveals their structure, consistency, and consequences.

Maxwell's equations are not only equations of electricity and magnetism; they are the equations that revealed light itself to be an electromagnetic wave.